

SHELLUI:

BUILDING WEBAPP FROM THE GROUND UP

VIBEHUUS

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TODAY'S TALK

WHAT TO EXPECT

- What ShellUI is – and the problem it solves
- The philosophy – why we built it this way
- Vibe-coding it – my experience shipping with AI
- Conclusion – takeaways and what's next

SÉBASTIEN BARBIER

ABOUT ME



Full-stack developer, more frontend – React & Python. Based in Zurich. I love building projects – even the ones that never go live or that no one uses.

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WHY SHELLUI?

FROM BUILDING TO SHIPPING

- Developers love to build – but struggle to ship
- AI changed that: agents let you turn ideas into code very quickly
- Every new product still reimplements the same general features
- Those features deserve a shared codebase – that's what I've been building with ShellUI

WHAT SHELLUI IS

THE PROBLEM IT SOLVES

DISCLAIMER

WHAT SHELLUI IS

- Today it's a proof of concept – a one-person project, still taking shape
- Experimental – expect breaking changes
- Not in production – nothing is battle-tested yet
- Aiming for a first release later this year – with serious ambition
- But it already looks pretty cool... I think

MICROFRONTEND SHELL

WHAT SHELLUI IS

ShellUI is a microfrontend – a wrapper around your app with an opinionated take on basic product features.

- The shell is built with React and shadcn/ui
- Your app runs in an iframe and communicates with the shell through the postMessage API
- Your embed can use any stack – Angular, React, plain HTML, ...

FEATURES

WHAT SHELLUI IS

NAVIGATION MENU

TOASTER

MODAL

DRAWER

LAYOUT

THEME

I18N

L10N

OFFLINE

COOKIE CONSENT

LEGALS

SOFTWARE UPDATE

AND MORE

Honestly? A two-minute demo beats this slide every time.

SHELL UI TODAY

WHAT SHELLUI IS

WEBSITE

`shellui.com`

DOCUMENTATION

`docs.shellui.com`

PLAYGROUND

`playground.shellui.com`

NPM PACKAGES

`@SHELLUI/CORE`

`@SHELLUI/CLI`

`@SHELLUI/SDK`

NEEDS A BACKEND

WHAT SHELLUI IS

- The shell is the UI – you still need a backend for real products
- Supabase is the popular default – ShellUI supports it out of the box
- We also ship a self-hosted Django backend for teams that need deeper, custom integrations

BACKEND APIS

WHAT SHELLUI IS

Shared product features as APIs – so every app does not rebuild the same plumbing.

AUTH

FILE STORAGE

LLM GATEWAY

FILE STORAGE & LLM GATEWAY – COMING SOON

Same rule as before – let's look at auth in the demo, not on this slide.

SHELL UI TODAY

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@SHELLUI/CLI

@SHELLUI/SDK

BACKEND & SERVICES

AUTH BACKEND

id.shellui.com

ADMIN PANEL

admin.shellui.com

FILES

COMING SOON

LLM GATEWAY

COMING SOON

ALL OPEN SOURCE – <https://github.com/shellui>

THE PHILOSOPHY

WHY WE BUILT IT THIS WAY

ARCHITECTURE

THE PHILOSOPHY

- Microservices + micro-frontends
- Every service exposes a clear, defined interface – I know what it does, and that's the only surface I need to care about
- Those APIs are versioned, tested, and not supposed to break

PROS

THE PHILOSOPHY

- Smaller codebases – easier to review and for AI context
- Limited blast radius when something goes wrong
- Deploy and iterate on one service without touching the rest
- Clear boundaries – fewer surprise side effects

CONS

THE PHILOSOPHY

- More moving parts – ops, networking, observability
- Harder to trace bugs across service boundaries
- Integration and contract testing take more work
- Some duplication – types, auth, shared UI patterns

SEPARATION OF CONCERNS

THE PHILOSOPHY

- Main goal: embrace clear boundaries between pieces
- Vibe-coding in one area shouldn't break code somewhere else
- Work inside a slice – frontend, API, or worker – with confidence
- Room for trash code in a slice – easy to drop when it's done

VIBE-CODING IT

SHIPPING WITH AI

VIBE-CODING IT

THE CODE

- Goal: spend time on the code, not on the AI
- One feature at a time
- Know the code structure – understand it all
- Review, test, and document every result – pretty aggressive micromanagement

VIBE-CODING IT

THE AI

- Cheap Cursor plan only – ~20 CHF/month
- Didn't write a single line – all done by prompting
- Even typos, color tweaks, and minor fixes went through the agent

ROOM FOR IMPROVEMENT

VIBE-CODING IT

- Still plenty to optimise in how I work with the agent
- Add skills – encode conventions and workflows for the project
- Add MCPs for backend services – safer context, fewer wrong guesses

CONCLUSION

TAKEAWAYS

- Not for everyone – strict process and real ownership of the code
- Not for every project – the extra structure has to earn its keep
- Still worth knowing: microservices + micro-frontends brought me peace of mind

THANK YOU

ANY QUESTIONS?

Looking for a developer? I'd love to hear about your team – let's talk about hiring me.

SLIDES AVAILABLE HERE



slides.sebastienbarbier.com